ESTIMATION OF ROOFTOP SOLAR PV SYSTEM (1 - 20 kW)

Date: 22-Sep-2026

Site: Case C - Deficit Area Customer | Location: Hubballi, Karnataka | Grid-Tied Solar Engineering Dossier

PLANT CAPACITY	ANNUAL YIELD	CAPITAL COST	PAYBACK PERIOD
550W Mono	Specific Yield	12-Item BOM	25Y Lifecycle
10.00 kW	**16,569 kWh**	**Rs. 447,450**	**4Y 5M**
10.45 kWp Installed	CUF 18.1% | PR 75%	Rs. 4.47 Lakhs	Profit: Rs. 49.12L

[SITE] Customer & Site Engineering Parameters

Verified project configuration, contact details, and sizing pathway

Company / Customer Name	Case C - Deficit Area Customer
Contact Mobile / Phone	+91 96666 33333
Location / Address	Hubballi, Karnataka
Contact Email Address	case_c@voltroof.com
Design Path Selected	Rooftop Area Analysis
Approved Sanctioned Load (CD)	20 kW (KERC / ESCOM Tariff Order)
Rooftop Area Verification	Deficit Rooftop Area (-800 sq.ft deficit, limited to 10.00 kW)
System Target Capacity	10.00 kW (Single LT Installation)
Total Installed Array Rating	10.45 kWp
Solar PV Module Rating	550W Mono PERC / TOPCon Half-Cut
Total Module Count	19 Nos.

[METRICS] System Performance & Economic Evaluation

25-Year performance metrics and grid benchmark comparisons

Total Rooftop Area	800.00 Sq.Feet (74.32 sq.m)
Utilized Usable Rooftop Area	800 Sq.Feet (74.32 sq.m)
First Year Annual Generation	16,568.88 kWh
25-Year Cumulative Generation	368,125.12 kWh
Estimated Payback Period	4 Years & 5 Months
Post-Payback Profitability (25 Yrs)	Rs. 4,912,278.67 (INR 49.12 Lakhs)
Alternative FD 6.5% Benchmark (25 Yrs)	Rs. 2,160,153.97 (INR 21.60 Lakhs)

ENVIRONMENTAL & SUSTAINABILITY IMPACT (25-YEAR LIFECYCLE)

Carbon Offset: 301 Tonnes CO2 | Tree Equivalence: 481 Mature Trees | Green Energy: 100% Clean LT Power

Based on standard Central Electricity Authority (CEA) grid emission and forestry carbon sequestration benchmarks.

20 kW Reached For LT Limited 20kW as per KERC/ESCOM Regulations

Statutory ceiling enforced under KERC SRTPV Regulations. System automatically sized to maximum allowable 20.0 kW.

[ARRAY] Building 1 Layout & Rooftop Array Sizing

Building structural envelope, array capacity, and module layout

Building / Array Identification	Building 1 (Rooftop Array)
Gross Rooftop Area	800.00 Sq.Feet (74.32 sq.m)
Utilized Effective Rooftop Area	800 Sq.Feet (74.32 sq.m)
Building Plant Target Capacity	10.00 kW
Solar PV Module Rating	550 W Mono-PERC (Mono-facial)
Number of Solar Modules	19 Nos.
String Sizing Architecture	18 - 20 Panels per MPPT String (Optimal Voltage Band)
Selected Site Inverter Rating	8 kW, 415V 3-Ph MPPT

[INVERTER] Selected Grid-Tied Inverter Specifications (8 kW)

Core inverter rating (70% derated selection), MPPT topology & models

Selected Inverter Rating	8 kW Grid-Tied Inverter (Three-Phase)
Certified Top-Tier OEM Models	GoodWe GW8K-DT, Polycab PSIT 8K
Derated AC Design Power (70% Factor)	7.31 kW AC (from 10.00 kWp DC)
AC Output Voltage & Grid Phase	415V, 3-Phase, 50 Hz AC
MPPT Operating Topology	Dual MPPT Architecture (>99.5% MPPT Tracking Efficiency)
Enclosure & Grid Protection	IP65 Weatherproof Enclosure with Anti-Islanding (IEC 62116)
Inverter to ACDB Cable	3.5Cx10 Sq.mm
DC Array Isolator Switch	32A, MCB Type, 4P 415V

[MCB] Selected MCB & Switchgear Protection Specifications

Breaker sizing, trip characteristics, and surge protection devices

Incomer Overcurrent Protection	Thermal-Magnetic MCB / MCCB with Manual Isolation
Breaker Current Rating	25 Amps
Rated Inverter Operating Current	10.17 Amps
Trip Curve & Breaking Capacity	Type C Curve / 10 kA Breaking Capacity (IS/IEC 60898-1)
Surge Protection Device (SPD)	Type II AC SPD (40 kA, 415V / 230V) with Remote Signalling

[CABLES] Electrical Cable Sizing & Interconnection Schedule

Cable conductor specifications, insulation, and voltage drop compliance

Solar PV DC Array Cable	4 / 6 sq.mm UV-Resistant Solar DC Cable (1.5 kV, EN 50618)
Inverter AC Output Cable	3.5Cx10 Sq.mm (Cu / XLPE / Armoured)
ACDB to Utility Grid Cable	3.5Cx6 Sq.mm (Al / XLPE / Armoured)
Earth Grounding Conductor	16 sq.mm Cu / GI Earth Conductor connected to Dual Chemical Pits
Design Standard & Voltage Drop	Voltage Drop < 2% in accordance with IS 694 / IEC 60364-5-52

[ACDB] AC Distribution Board (ACDB 1 IN / 1 OUT) Specifications

Switchgear enclosure, busbar rating, and metering point

ACDB Enclosure Specification	IP65 Weatherproof Polycarbonate / Sheet Steel
Internal Busbar Rating	50A, 415V, 3PN Aluminium
Incomer & Outgoing Terminations	Cable Glands with Compression Lugs & Dual Earthing
Interconnection Distance	20.0 Meters (ACDB to Utility Net-Meter)

[BOM] Turnkey Plant Bill of Materials (BOM)

Complete 12-item engineering specification and cost breakdown

20 kW Reached For LT Limited 20kW as per KERC/ESCOM Regulations

Sl.	Component / Material Description	Engineering Specification	Qty	Amount (INR)
01	Solar PV modules	550W Mono-PERC (Mono-facial)	19 Nos.	Rs. 190,000
02	On-grid inverter	8 kW, three-phase (GoodWe GW8K-DT or equiv.)	1 No.	Rs. 88,000
03	Rooftop mounting structure	HDGI, module-compatible	1 Set	Rs. 45,000
04	DC solar cable	6 sq.mm copper, PV-rated	110 m	Rs. 9,350
05	AC cable	16 sq.mm copper	60 m	Rs. 14,400
06	MC4 connectors	PV-rated connector pairs	19 pairs	Rs. 5,700
07	DCDB	Isolator + fuse + Type-II DC SPD	1 No.	Rs. 11,000
08	ACDB	25A MCB + 40A RCCB + Type-II AC SPD	1 No.	Rs. 11,500
09	Earthing kit	Earth electrodes & accessories (IS 3043 - 3 Pits)	3 Nos.	Rs. 7,500
10	Lightning arrester	Standard rooftop PV type	1 No.	Rs. 5,000
11	Installation accessories	Lugs, ferrules, clamps, conduit, glands, etc.	1 Lot	Rs. 15,000
12	Installation & commissioning	Labour and commissioning	1 Job	Rs. 45,000
		TOTAL PLANT-SIDE CAPITAL COST (12-ITEM BOM)		**Rs. 447,450.00**

[UTILITY] ESCOM / Utility Statutory Approvals & Grid Budget

Official DISCOM statutory filing fees and net-metering charges

Application & Registration Fee	Rs. 1,000 + GST (DISCOM portal filing)
Facilitation & Technical Processing Fee	Rs. 2,000 + GST (Feasibility clearance)
PPA e-Sign Stamp Duty Charge	Rs. 11.8 (Electronic stamping)
Bidirectional Utility Net-Meter	Rs. 12,000 (DISCOM meter testing/procurement)
Refundable Security Deposit	Rs. 8,000 (Refundable utility escrow)

TOTAL CAPITAL INVESTMENT REQUIRED

Turnkey Plant BOM Cost: Rs. 447,450 | ESCOM Utility Budget: Rs. 23,012 | Grand Total Outlay: Rs. 470,462 (Rs. 4.70 Lakhs)

[FINANCE] 25-Year Financial Amortization Schedule

Annual cash flow model incorporating 1%/year degradation & tariff escalation

CAPITAL INVESTMENT (CAPEX)	PAYBACK PERIOD (RETURN)	25Y NET SAVINGS (PROFIT)	FD 6.5% BENCHMARK (BENCHMARK)
Rs. 447,450	**4Y 5M**	**Rs. 49.12L**	**Rs. 21.60L**
12-Item BOM Basis	Break-Even Year	Rs. 4,912,279	Alternative Asset

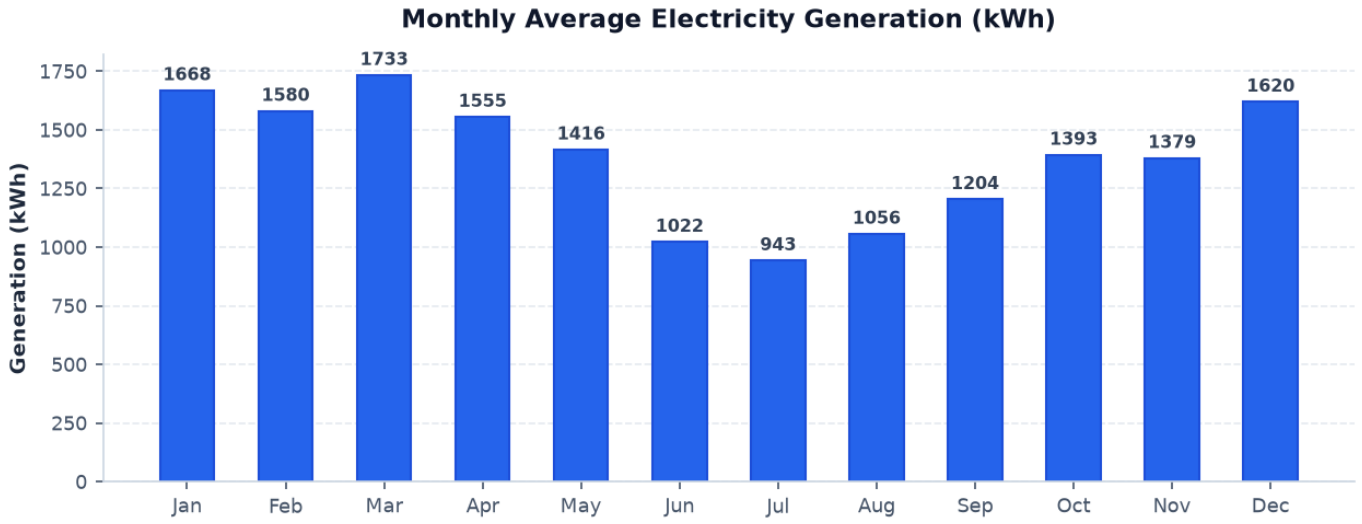
Yr	Gen (kWh)	Tariff (Rs/u)	Gross Rev (Rs)	AMC (Rs)	Net Annual (Rs)	Cumulative Net (Rs)
01	16,569	Rs. 8.05	133,379	6,000	127,379	127,379
02	16,403	Rs. 8.45	138,607	6,000	132,607	259,986
03	16,239	Rs. 8.87	144,041	6,000	138,041	398,028
04 *	**16,077**	**Rs. 9.32**	**149,835**	**6,000**	**143,835**	**541,863**
05	15,916	Rs. 9.78	155,658	6,000	149,658	691,522
06	15,757	Rs. 10.27	161,823	6,000	155,823	847,344
07	15,599	Rs. 10.78	168,160	6,000	162,160	1,009,505
08	15,443	Rs. 11.32	174,818	6,000	168,818	1,178,323
09	15,289	Rs. 11.89	181,784	6,000	175,784	1,354,107
10	15,136	Rs. 12.48	188,897	6,000	182,897	1,537,004
11	14,985	Rs. 13.11	196,448	6,000	190,448	1,727,452
12	14,835	Rs. 13.76	204,126	6,000	198,126	1,925,578
13	14,686	Rs. 14.45	212,219	6,000	206,219	2,131,797
14	14,540	Rs. 15.18	220,710	6,000	214,710	2,346,507
15	14,394	Rs. 15.93	229,299	6,000	223,299	2,569,805
16	14,250	Rs. 16.74	238,548	6,000	232,548	2,802,354
17	14,108	Rs. 17.57	247,872	6,000	241,872	3,044,226
18	13,967	Rs. 18.45	257,684	6,000	251,684	3,295,910
19	13,827	Rs. 19.37	267,828	6,000	261,828	3,557,739
20	13,689	Rs. 20.34	278,428	6,000	272,428	3,830,167
21	13,552	Rs. 21.36	289,466	6,000	283,466	4,113,633
22	13,416	Rs. 22.43	300,927	6,000	294,927	4,408,560
23	13,282	Rs. 23.55	312,794	6,000	306,794	4,715,354
24	13,149	Rs. 24.73	325,182	6,000	319,182	5,034,536
25	13,018	Rs. 25.96	337,942	6,000	331,942	5,366,479

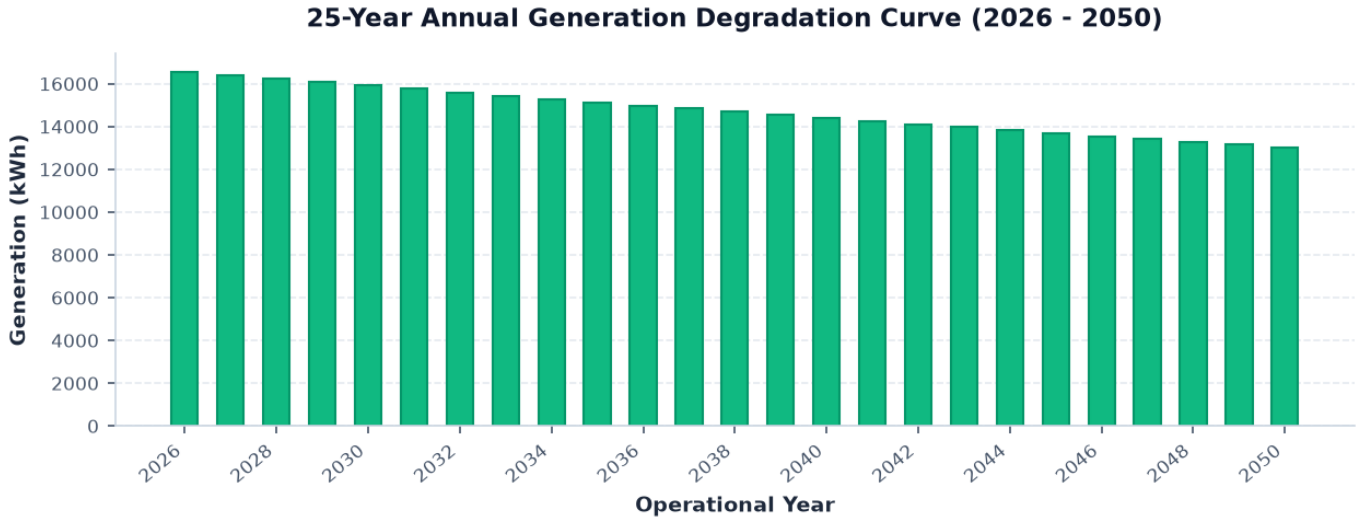
** Highlighted row indicates capital payback year. Annual maintenance contract (AMC) budgeted at Rs. 600/kW/year with 1% annual PV degradation.*

[ANALYTICS] Generation Analytics & Performance Curves

Monthly solar insulation yield and 25-year degradation profile

SPECIFIC YIELD (YIELD)	PERFORMANCE RATIO (PR)	CAPACITY UTIL. (CUF)	DEGRADATION RATE (DEGRADATION)
1657 kWh/kWp	**75.0%**	**18.1%**	**1.0% / year**
kWh/kWp/Year	IEC 61724 Standard	Annual Plant Factor	Tier 1 Linear Warranty





Simulated with regional GHI solar insolation database (3.26 to 5.40 kWh/m2/day). System performance calculated using PVsyst/MNRE benchmarks.

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CLEAN ENERGY. BRIGHTER TOMORROW

ON-GRID SOLAR PV SYSTEM

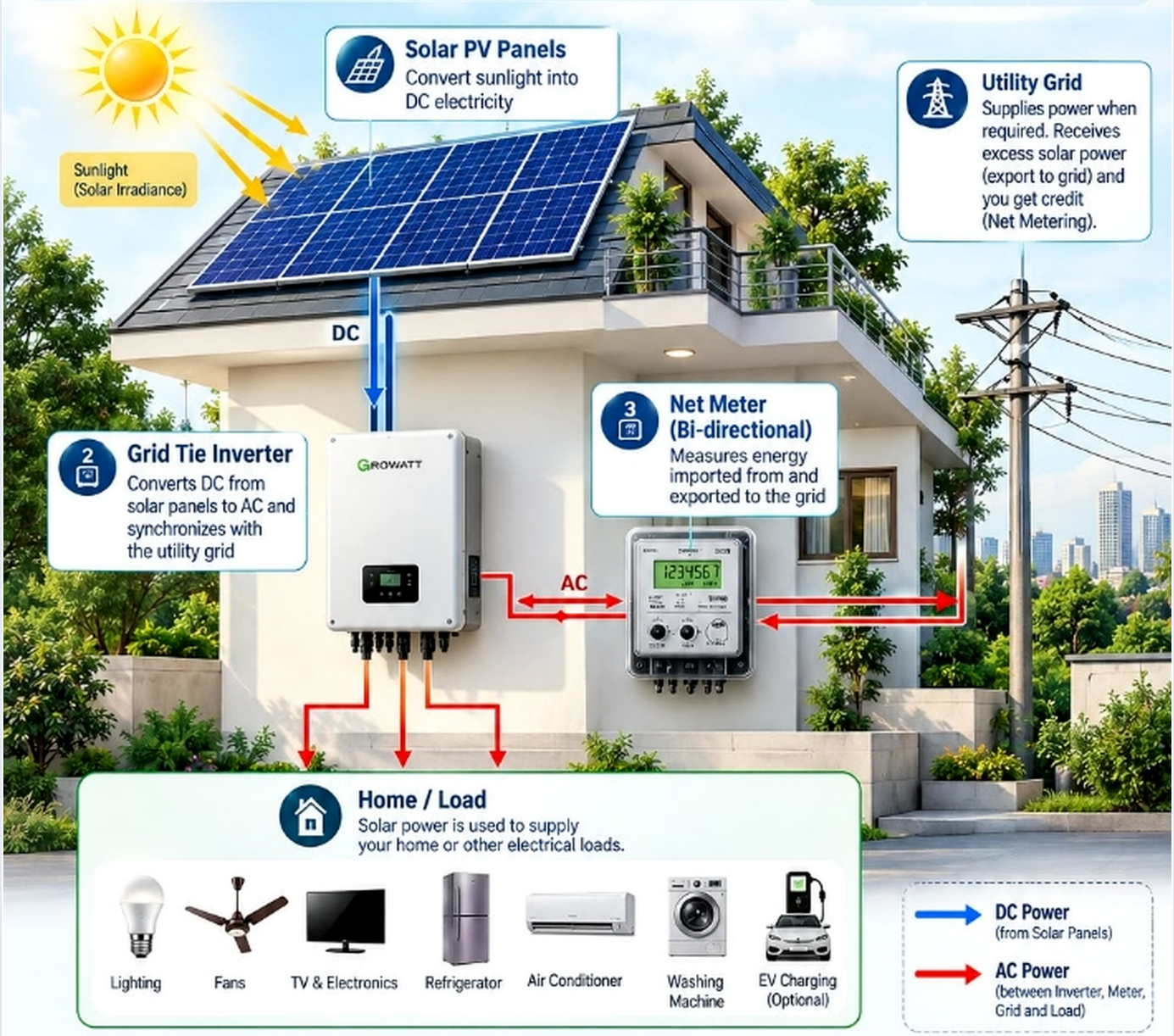
(Grid Connected Solar System)

GENERATE • CONSUME • EXPORT • SAVE

Harness The Sun For A Better Tomorrow

An on-grid solar PV system is connected to the utility grid. The solar power generated is used to meet your load demand. Any excess power is exported to the grid and you get credit through net metering. No battery is used in this system.

- Lower Electricity Bills
- Clean Energy
- Sustainable Future



Home / Load

Solar power is used to supply your home or other electrical loads.

Lighting | Fans | TV & Electronics | Refrigerator | Air Conditioner | Washing Machine | EV Charging (Optional)

DC Power (from Solar Panels)

AC Power (between Inverter, Meter, Grid and Load)

KEY ADVANTAGES

- Reduced Electricity Bills
- Lower Carbon Emissions
- No Battery Required
- Net Metering Benefits
- Low Maintenance
- Environment Friendly

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Clean Energy Better Tomorrow